

MOHAMMED YAHYA

Kalaburagi, Karnataka, India 585104

9481292430 yahya94812@gmail.com [LinkedIn](#) [GitHub](#) [HuggingFace](#) [Portfolio](#)

Education

PDA College of Engineering

Bachelor of Engineering in Computer Science

Aug 2023 – July 2027

Kalaburagi, Karnataka

Relevant Coursework

- Large Language Models (LLMs)
- LLM Pretraining & Fine-tuning (LoRA & QLoRA)
- PyTorch & Hugging Face Transformers
- GPU Training Optimization
- Applied Artificial Intelligence
- Machine Learning Operations (MLOps)

Projects

Tiny-GPT | [Hugging Face](#)

PyTorch, Transformer Architecture, Flash Attention, CUDA/BF16, Hugging Face, Matplotlib

July 2026

- Built a 0.1B-parameter GPT-style language model from scratch in PyTorch, implementing the full transformer stack — multi-head self-attention with fused QKV projections, Flash Attention via `scaled_dot_product_attention`, weight-tied embeddings, and a GELU-based MLP block — trained on ~2B tokens for \$12 in 3 hours on a single H100 GPU.
- Engineered a production-grade training loop with gradient accumulation (effective batch size ~524K tokens), mixed-precision BF16 autocast, gradient clipping, and a custom linear-warmup/cosine-decay learning rate schedule, alongside a multithreaded data loader that prefetches shards in the background to eliminate I/O stalls.
- Preprocessed and published a cleaned, tokenized dataset and trained model weights to Hugging Face Hub, enabling one-command inference and training reproduction; the model learned coherent grammar and long-range context, generating fluent short stories from raw character-level input.

Cogni-Agent | [Hugging Face](#)

PyTorch, Hugging Face TRL/PEFT, QLoRA, BitsAndBytes, Web Search API

October 2025 – January 2026

- Fine-tuned Microsoft's Phi-4-mini-instruct (4B) with 4-bit QLoRA to reduce reliance on parametric memory, instead teaching the model to reason step-by-step and issue live web-search tool calls to retrieve up-to-date information rather than recalling static, potentially outdated facts from its weights.
- Trained on the RUC-AIBOX/SimpleDeepSearcher dataset using Hugging Face's SFT Trainer with a custom chat template, applying loss masking over web-search-result tokens so the model learned when and how to search rather than memorizing or hallucinating retrieved content.
- Built a deployment loop that intercepts model-issued search queries at inference time, executes them against the DuckDuckGo Search API, and re-injects results into the model's context window — enabling grounded, in-context-learning-driven responses instead of static knowledge retrieval.

Prompt-Router | [Hugging Face](#)

ModernBERT, PyTorch, Hugging Face Transformers, Synthetic Data Generation

February 2025 – April 2025

- Trained a ModernBERT-based classifier from scratch to route incoming LLM prompts across categories (medical, coding, learning, emotional support, planning, etc.), enabling intelligent dispatch to task-specialized LLMs with per-category configs — temperature, injected skills, and output format — instead of a single one-size-fits-all model call.
- Generated a synthetic labeled training corpus by using GPT-OSS-20B to classify and annotate prompts from the `fka/prompts.chat` dataset across domain and reasoning-effort dimensions (e.g., factual lookups tagged low-reasoning/low-temperature vs. medical or planning queries tagged high-reasoning), producing category- and difficulty-aware ground truth at scale.
- Wrapped the trained classifier in a FastAPI service that routes each prompt to the optimal downstream LLM and dynamically injects category-specific skills/context, cutting unnecessary high-cost model usage by matching prompt complexity to the cheapest sufficient model — reducing inference spend for business use cases.

Technical Skills

Frameworks & Libraries: PyTorch, Hugging Face Transformers, TRL (SFTTrainer), PEFT, Matplotlib

Developer Tools: Colab, Kaggle, W&B, Hugging Face, Lightning AI, llama.cpp, Serverless GPUs

Technologies: Transformers, LoRA/QLoRA, Synthetic Data Generation, Custom Training Loops, Inference Optimization

Academic Results

- Scored 98/100 in SSLC (10th Std)
- Scored 94/100 in PUC (12th Std)
- Secured Rank 810 out of 250,000+ in KCET
- Maintaining 8.6 CGPA through 6th semester of BE